Technology Stack

Date	03 October 2026
Team ID	2
Project Name	FitBuddy – AI Fitness Plan Generator using Gemini Models
Maximum Marks	2 Marks

Define Your Technology Stack:

FitBuddy uses the stack specified in the SmartInternz project brief, updated to current, supported versions.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3 (Flexbox, media queries), Jinja2 templates, Google Fonts (Roboto)	Required by the project brief; server-rendered pages need no JS framework, load fast and work on any device. Jinja2 auto-escaping protects against injected HTML.
2	Backend / Server-Side	Python 3.11+, FastAPI, Uvicorn (ASGI), Pydantic v2	FastAPI gives typed routes, automatic validation and free Swagger docs at /docs; Uvicorn is a fast ASGI server; Pydantic validates both user input and the AI's JSON.
3	Database / Data Storage	SQLite (fitbuddy.db, WAL mode) via SQLAlchemy 2.x ORM	Zero-setup file database, ideal for local deployment; ORM keeps the code portable to PostgreSQL. WAL mode prevents 'database is locked' under concurrent writes.
4	Cloud / Hosting / Deployment	Local deployment with Uvicorn; configuration through .env (python-dotenv)	The project brief targets local deployment. The app is stateless apart from the DB, so it can move to Cloud Run or Render later.
5	Version Control & CI/CD	Git, GitHub, pytest (245 tests), Locust (load testing)	GitHub hosts the code and phase deliverables; pytest verifies every scenario in demo mode; Locust produced the performance-test evidence.

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6	Third-Party APIs / Other Tools	Google Gemini API via google-genai SDK: gemini-3.8-flash (plans, revisions), gemini-3.5-flash-lite (tips, fallback)	Structured JSON output with a response schema gives reliable 7-day plans; two model roles balance quality and speed. The brief's Gemini 1.5 models are retired, so current models were used.

Key libraries (requirements.txt)

fastapi, uvicorn, jinja2, sqlalchemy, python-multipart, python-dotenv, google-genai, itsdangerous (signed cookies), pytest, httpx, locust.